Lecture 5: Likelihoods, Bayes' theorem, and the Laplace approximation

COMP 521 • Probability, Optimization, and Scientific Discovery • Sections 3.1–3.4

This handout is one part of the full COMP 521 notes. Section, equation, figure and listing numbers match the full notes, and references to anything outside this handout point there. The notation table is in "How to use these notes" (Lecture 1).

3 From optimization to statistical inference

3.1 A likelihood is not a probability of the parameters

Given data $\mathbf{d}$, the *likelihood* is the sampling model for the data, read as a function of the parameters:

$$\boldsymbol{\theta} \mapsto p(\mathbf{d} \mid \boldsymbol{\theta}), \qquad \ell(\boldsymbol{\theta}) = \log p(\mathbf{d} \mid \boldsymbol{\theta}). \tag{22}$$

It is a density in $\mathbf{d}$, not in $\boldsymbol{\theta}$: integrating it over $\boldsymbol{\theta}$ does not generally give one, and it has no meaning as "the probability that $\boldsymbol{\theta}$ is true". Its negative logarithm is a loss, and minimizing it gives the *maximum-likelihood estimate* $\hat{\boldsymbol{\theta}}_{\mathrm{ML}} = \arg\max_{\boldsymbol{\theta}} p(\mathbf{d} \mid \boldsymbol{\theta}) = \arg\min_{\boldsymbol{\theta}}[-\ell(\boldsymbol{\theta})]$.

The MLE has a useful invariance: if $\boldsymbol{\varphi} = g(\boldsymbol{\theta})$ is a reparameterization, then $\hat{\boldsymbol{\varphi}}_{\mathrm{ML}} = g(\hat{\boldsymbol{\theta}}_{\mathrm{ML}})$, because a likelihood value is attached to a parameter *point* and carries no Jacobian. The MLE is a point; it says nothing about uncertainty on its own.

3.2 Bayes' theorem

The product rule written both ways, $p(\boldsymbol{\theta}, \mathbf{d}) = p(\mathbf{d} \mid \boldsymbol{\theta})p(\boldsymbol{\theta}) = p(\boldsymbol{\theta} \mid \mathbf{d})p(\mathbf{d})$, gives

$$\underbrace{p(\boldsymbol{\theta} \mid \mathbf{d})}_{\text{posterior}} = \frac{\overbrace{p(\mathbf{d} \mid \boldsymbol{\theta})}^{\text{likelihood}}\ \overbrace{p(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(\mathbf{d})}_{\text{evidence}}}, \qquad p(\mathbf{d}) = \int p(\mathbf{d} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\theta}. \tag{23}$$

The posterior is a full distribution over parameters. Its mode is the *maximum a posteriori* (MAP) estimate, $\boldsymbol{\theta}_{\mathrm{MAP}} = \arg\min_{\boldsymbol{\theta}}[-\ell(\boldsymbol{\theta}) - \log p(\boldsymbol{\theta})]$, which is exactly the regularized fit of Section 2.7.

The MAP depends on your coordinates

A posterior *density* transforms with the Jacobian of Eq. (10), so its peak moves when you change variables. Example: if $p(\theta \mid \mathbf{d}) \propto \theta e^{-\theta}$ for $\theta > 0$, the MAP is $\theta = 1$. In $z = \log\theta$ the density is $p(z \mid \mathbf{d}) \propto \theta^2 e^{-\theta}$, whose peak is at $\theta = 2$. The posterior *probability* of any interval is the same in both coordinates; only the summary changed. Report posterior medians and quantile-based (equal-tailed) intervals, which do transform consistently, when the parameterization is arbitrary.

3.2.1 The linear-Gaussian model, completely

Let $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$ and $y = a\theta + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma^2)$. The log posterior is a quadratic in θ,

$$\log p(\theta \mid y) = -\frac{(\theta - \mu_0)^2}{2\tau_0^2} - \frac{(y - a\theta)^2}{2\sigma^2} + \text{const},$$

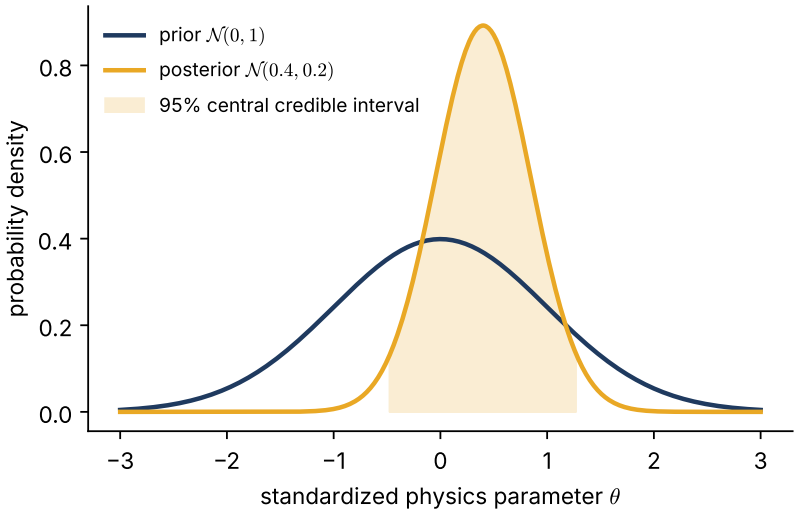


Figure 4: Bayesian updating in the linear-Gaussian TOY model: the posterior is narrower than the prior and shifted toward the value the data favour.

and completing the square gives a Gaussian posterior whose *precisions add*:

$$\frac{1}{\tau_{\text{post}}^2} = \frac{1}{\tau_0^2} + \frac{a^2}{\sigma^2}, \qquad \mu_{\text{post}} = \tau_{\text{post}}^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{a\,y}{\sigma^2} \right). \tag{24}$$

The posterior mean is a precision-weighted average of the prior mean and the value y/a implied by the data. For a vector $\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ and $\mathbf{y} = \mathbf{J}\boldsymbol{\theta} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$:

$$\boldsymbol{\Sigma}_{\text{post}} = \left(\boldsymbol{\Sigma}_0^{-1} + \mathbf{J}^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \mathbf{J} \right)^{-1}, \qquad \boldsymbol{\mu}_{\text{post}} = \boldsymbol{\Sigma}_{\text{post}} \left(\boldsymbol{\Sigma}_0^{-1} \boldsymbol{\mu}_0 + \mathbf{J}^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \mathbf{y} \right). \tag{25}$$

There is $\mathbf{J}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\mathbf{J}$ again, this time as the precision the data contribute.

Worked example 3.1 (Updating a standardized opacity correction)**.** Let $\mu_0 = 0$, $\tau_0 = 1$, $a = 2$, $\sigma = 1$, and observe $y = 1$. Then $1/\tau_{\text{post}}^2 = 1{+}4 = 5$, so $\tau_{\text{post}}^2 = 0.2$, and $\mu_{\text{post}} = 0.2{\times}(0{+}2{\times}1/1) = 0.4$:

$$\theta \mid y = 1 \sim \mathcal{N}(0.4,\, 0.2).$$

The standard deviation fell from 1 to $\sqrt{0.2} \approx 0.447$. The central 95% credible interval is $0.4 \pm 1.96\sqrt{0.2} \approx [-0.48, 1.28]$.

A *credible interval* contains a stated posterior probability, conditional on the model and prior. A frequentist *confidence interval* is produced by a procedure that covers the true value in a stated fraction of repeated experiments. They coincide in the linear-Gaussian model with a flat prior and differ in general; Section 3.7 shows how to test whether a Bayesian procedure has frequentist coverage.

3.3 The Laplace approximation

Most posteriors are not Gaussian, but many are nearly Gaussian near their peak. Expand the negative log posterior $U(\boldsymbol{\theta}) = -\ell(\boldsymbol{\theta}) - \log p(\boldsymbol{\theta})$ to second order about the MAP:

$$p(\boldsymbol{\theta} \mid \mathbf{d}) \approx \mathcal{N}\!\left(\boldsymbol{\theta}_{\text{MAP}},\, \mathbf{H}_U^{-1}\right), \qquad \mathbf{H}_U = \nabla^2 U(\boldsymbol{\theta}_{\text{MAP}}) \approx \mathbf{J}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\mathbf{J} + \boldsymbol{\Sigma}_0^{-1} \tag{26}$$

for Gaussian noise and a Gaussian prior, using the Gauss–Newton approximation of Eq. (17). This is what Listing 3 computed, with one shortcut: it centred the Gaussian on the maximum-likelihood fit rather than on the MAP. With $\mathcal{N}(0,1)$ priors and a data precision in the thousands, the two differ by far less than the posterior width. It costs one optimization plus one Hessian, and it fails for skewed, bounded, or multimodal posteriors, in which case you sample (Section 3.5).

One matrix, four jobs

For Gaussian noise with fixed covariance, the single matrix $\mathbf{J}^\mathsf{T}\boldsymbol{\Sigma}^{-1}\mathbf{J}$ is

1. the **Gauss–Newton Hessian** that sets optimizer steps (Eq. (17));
2. the **precision added by the data** in the linear-Gaussian posterior (Eq. (25)) and the Laplace approximation (Eq. (26));
3. the **Fisher information**, whose inverse bounds estimator variance (Section 4.2);
4. the matrix that sets the **expected information gain**, through $\det(\mathbf{I}+\boldsymbol{\Sigma}_0\mathcal{I})$, of a design (Section 4.5).

If you understand this matrix for your problem, you understand how hard it is to fit, how well it is constrained, and what would constrain it better.

3.4 Prediction, nuisance parameters, and evidence

To predict a new observable $\tilde{\mathbf{d}}$, average over what you do not know:

$$p(\tilde{\mathbf{d}} \mid \mathbf{d}) = \int p(\tilde{\mathbf{d}} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta} \mid \mathbf{d})\, \mathrm{d}\boldsymbol{\theta}. \tag{27}$$

This *posterior predictive distribution* combines observational scatter with parameter uncertainty; plugging in $\boldsymbol{\theta}_{\mathrm{MAP}}$ keeps only the first and understates the spread.

A *nuisance parameter* is needed by the data model but is not the scientific target. To constrain an exotic energy-loss coupling g, the conventional opacity, mixing, age, composition and calibration parameters $\boldsymbol{\phi}$ are nuisances. Bayesian analysis *marginalizes* them, $p(g \mid \mathbf{d}) = \int p(g, \boldsymbol{\phi} \mid \mathbf{d})\, \mathrm{d}\boldsymbol{\phi}$. Frequentist analysis often *profiles*:

$$\ell_{\mathrm{prof}}(g) = \max_{\boldsymbol{\phi}} \ell(g, \boldsymbol{\phi}). \tag{28}$$

The two answer different questions and give different intervals when the posterior in $\boldsymbol{\phi}$ is skewed or its width depends on g.

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A constraint on axion-like cooling from TRGB luminosities means something only if the conventional stellar uncertainties, observational error, selection, and competing explanations are in the model as nuisance parameters. A best-fitting nonzero coupling, on its own, is not evidence for a new particle.

The evidence and its prior sensitivity. The evidence $p(\mathbf{d})$ normalizes the posterior and compares models through Bayes factors $p(\mathbf{d} \mid \mathcal{M}_1)/p(\mathbf{d} \mid \mathcal{M}_2)$. It is much more sensitive to the prior than the posterior is. For a parameter with a uniform prior of width W and a likelihood of width $\sigma_L \ll W$, $p(\mathbf{d}) \approx p(\mathbf{d} \mid \hat{\theta})\sqrt{2\pi}\,\sigma_L/W$. So if $\mathcal{M}_2$ adds this parameter to $\mathcal{M}_1$, doubling its arbitrary prior width halves $p(\mathbf{d} \mid \mathcal{M}_2)$ and doubles the Bayes factor in favour of the simpler $\mathcal{M}_1$, although

the fit is unchanged. Bayes factors computed with vague priors therefore need prior-sensitivity checks.

Exercises for this lecture

Exercises 10, 11, numbered as in Appendix A of the full notes. Worked solutions are in Appendix B; attempt each one before reading its solution.

10. [P] Two measurements, one posterior. Prior $\theta \sim \mathcal{N}(0, 1)$; data $y_1 = \theta + \epsilon_1$ ($\sigma_1 = 1$, observed 0.8) and $y_2 = 3\theta + \epsilon_2$ ($\sigma_2 = 2$, observed 1.5). Find the posterior. Show that updating on y_1 and then on y_2 gives the same answer as updating on both at once.

11. [P] Moving MAP, fixed median. Let $p(\theta \mid \mathbf{d}) \propto \theta^{k-1} e^{-\theta/s}$ on $\theta > 0$. Find the MAP in θ and in $z = \log \theta$. For $k = 3$, $s = 1$, evaluate both. Which summary is invariant, and why?